

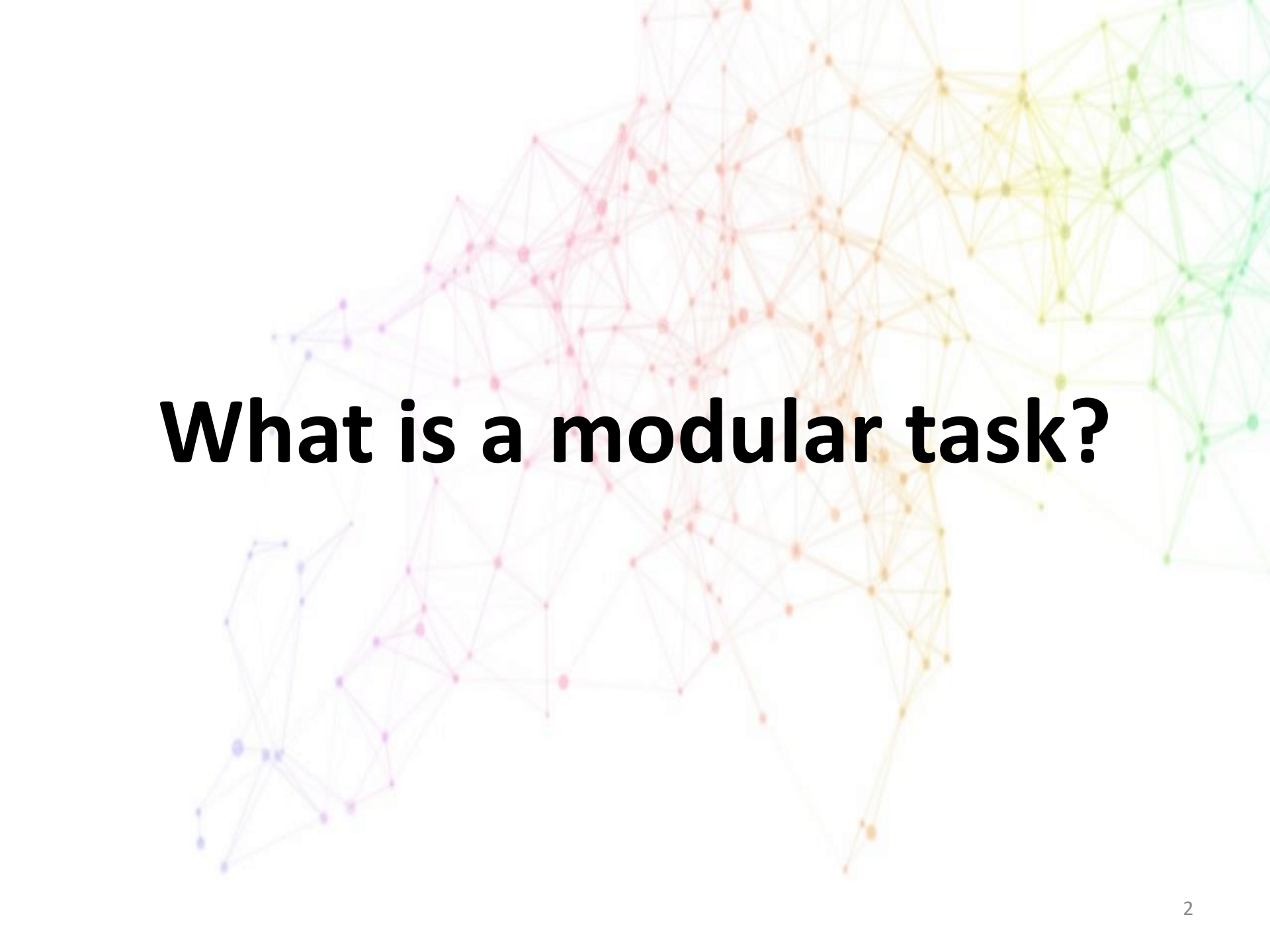
The Emergence of Modularity in Neural Networks

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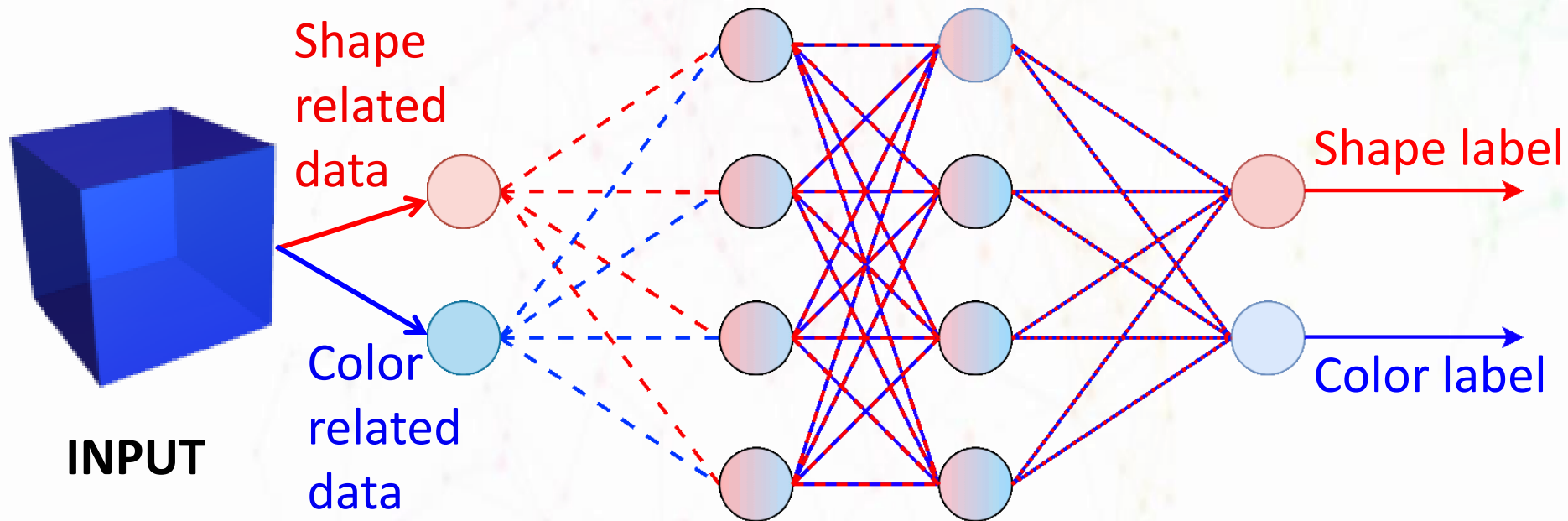
PI: Ila Fiete

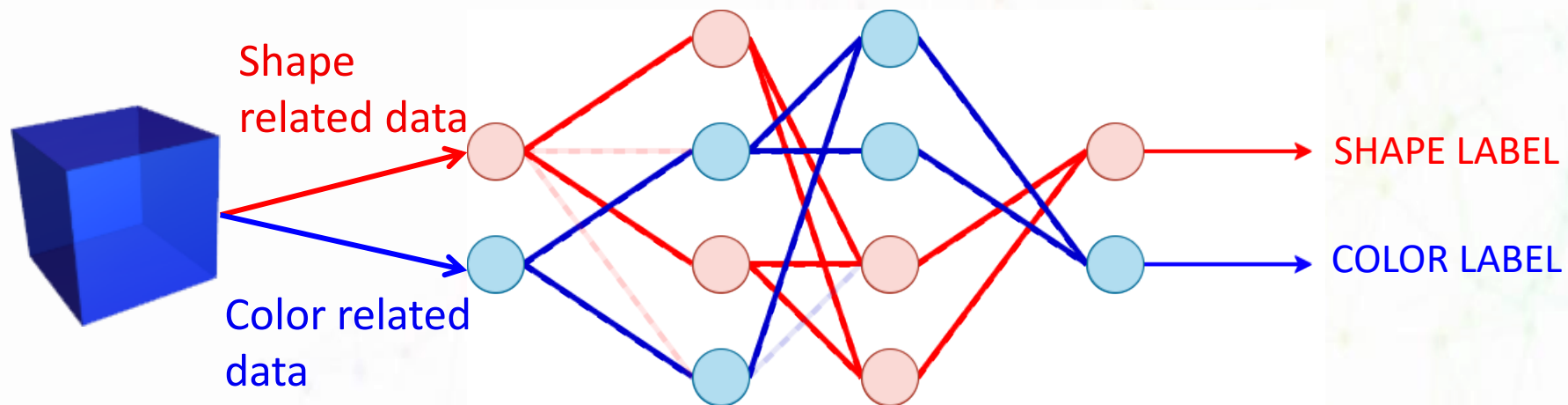




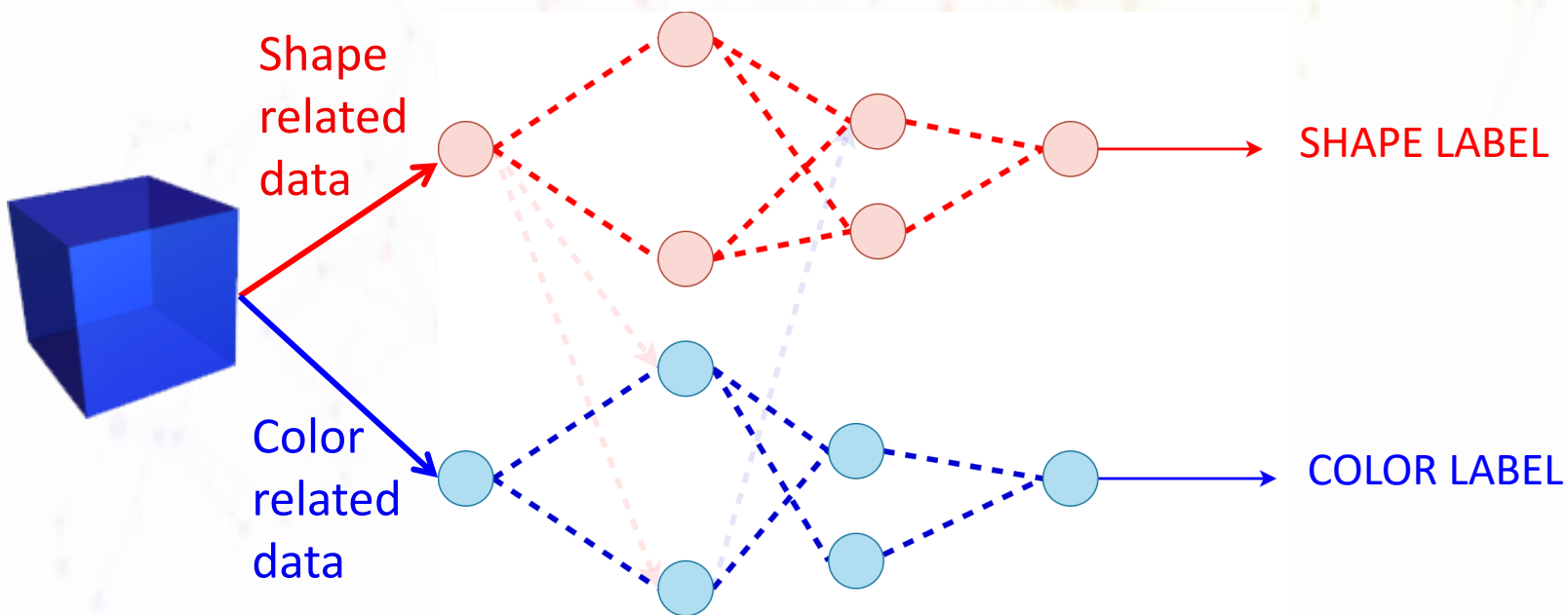
What is a modular task?

What is a modular neural network?





MODULAR NETWORK

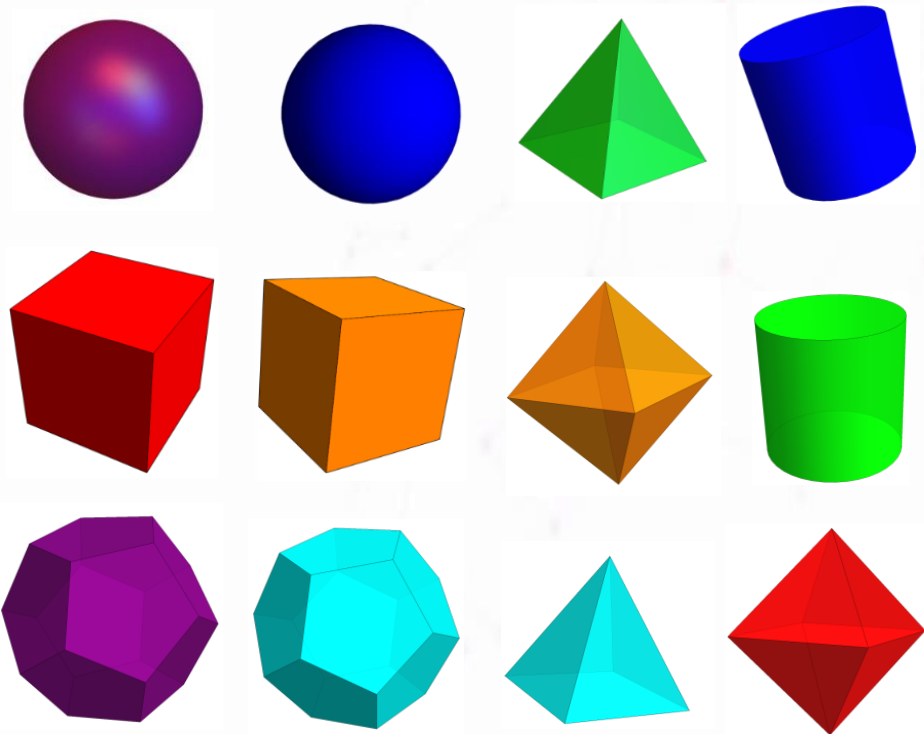


(same network)

Why modularity is important

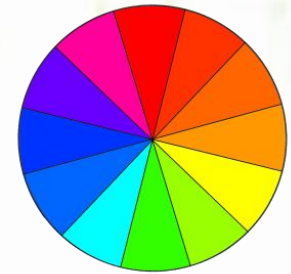
TASK = MODULAR
SYSTEM $\neq$ MODULAR

MANY UNIQUE COMBINATIONS TO
LEARN

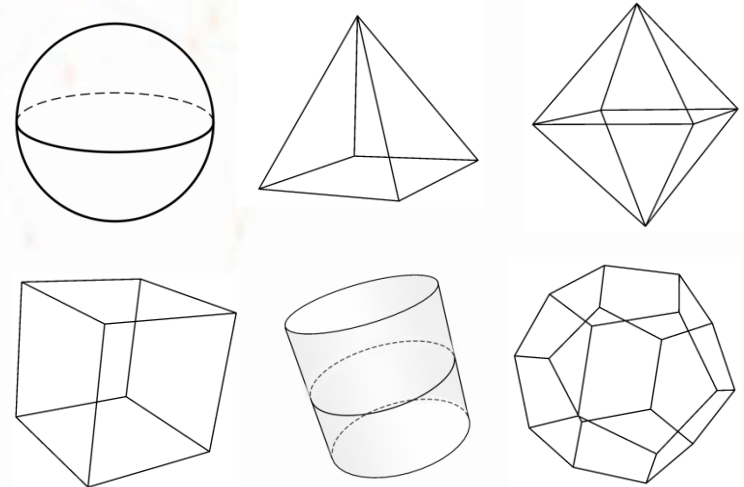


BUT, IF SYSTEM IS
MODULAR...

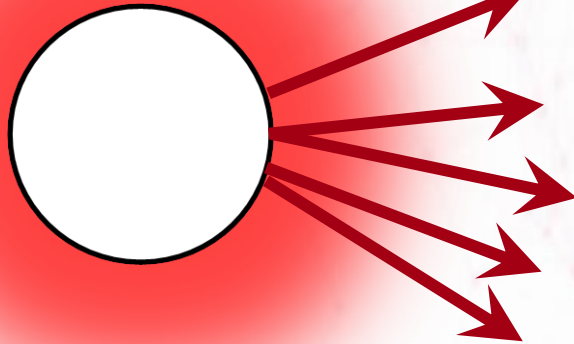
ONE
COMPONENT
LEARNS:



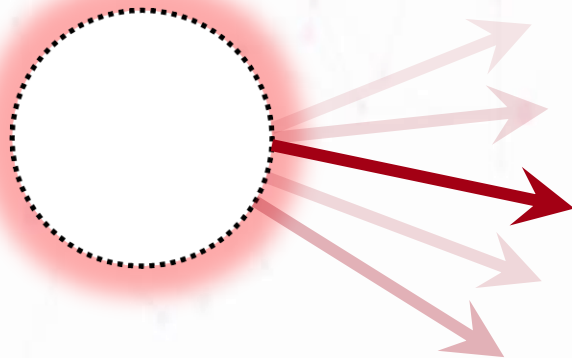
ANOTHER COMPONENT LEARNS:



How we encouraged modularity in neural networks



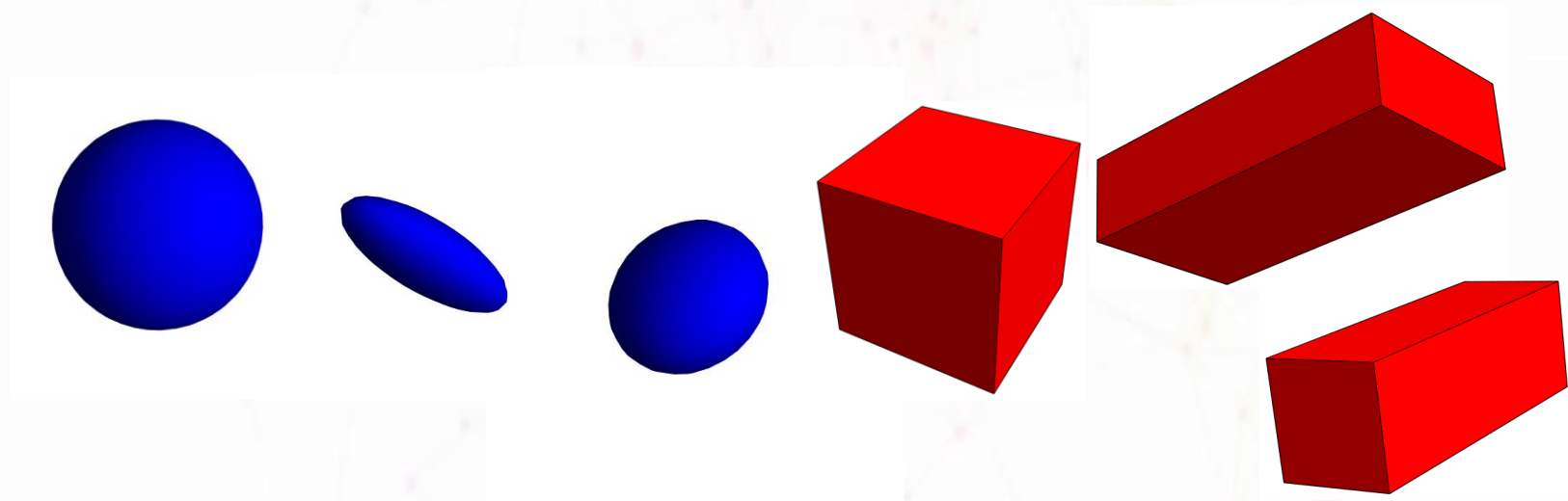
Highly active unit
+
Numerous/strong
connections
=
Large penalty



Weakly active unit
+
Few/weak
connections
=
Small penalty

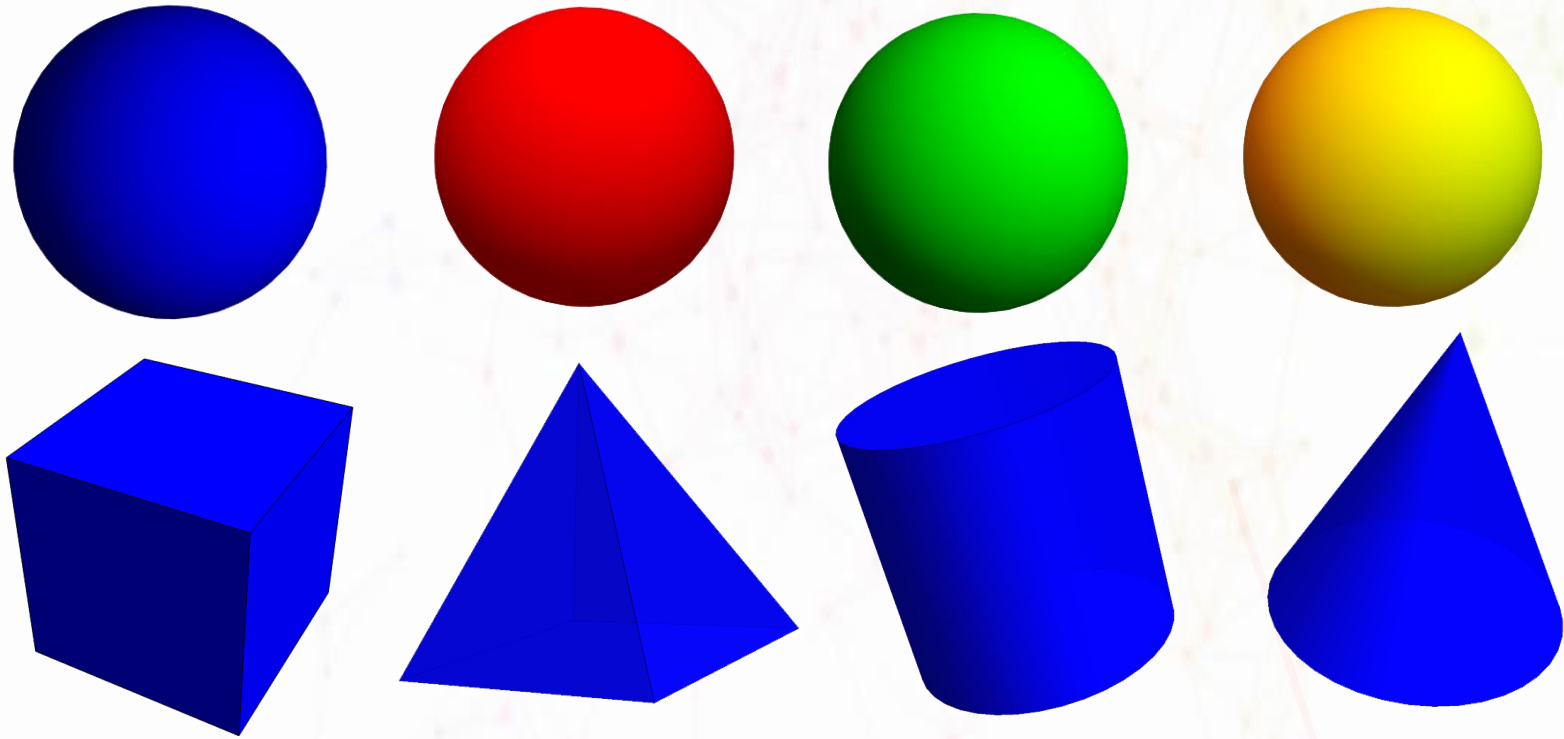
How does dataset design affect modularity in neural networks?

- Highly correlated properties → Poor modularity



- However, minimizing correlation between properties doesn't guarantee modular solutions

Highly contrasted combinations → Great modularity



- Hard to quantify exactly; may require a custom metric
- Best way to exploit this remains to be seen

Takeaways

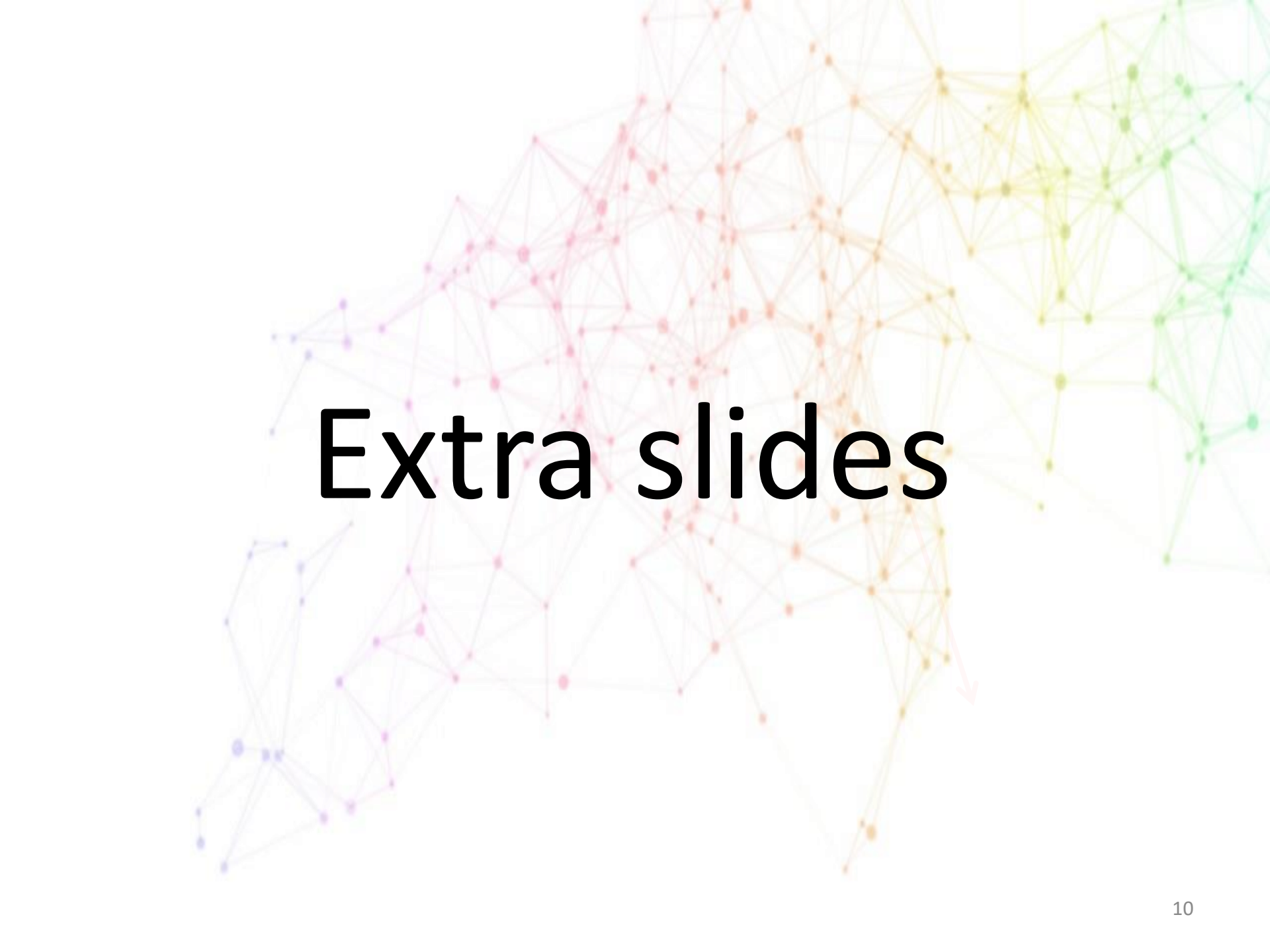
- Whether a network discovers a **modular solution depends on the data** it is trained on.
- **Strong statistical dependence inhibits modularity**, but reducing it does not guarantee it.
- **Repeated contrasts are highly effective** at inducing modularity.

Future work may examine how to measure and exploit this structural property for dataset construction.

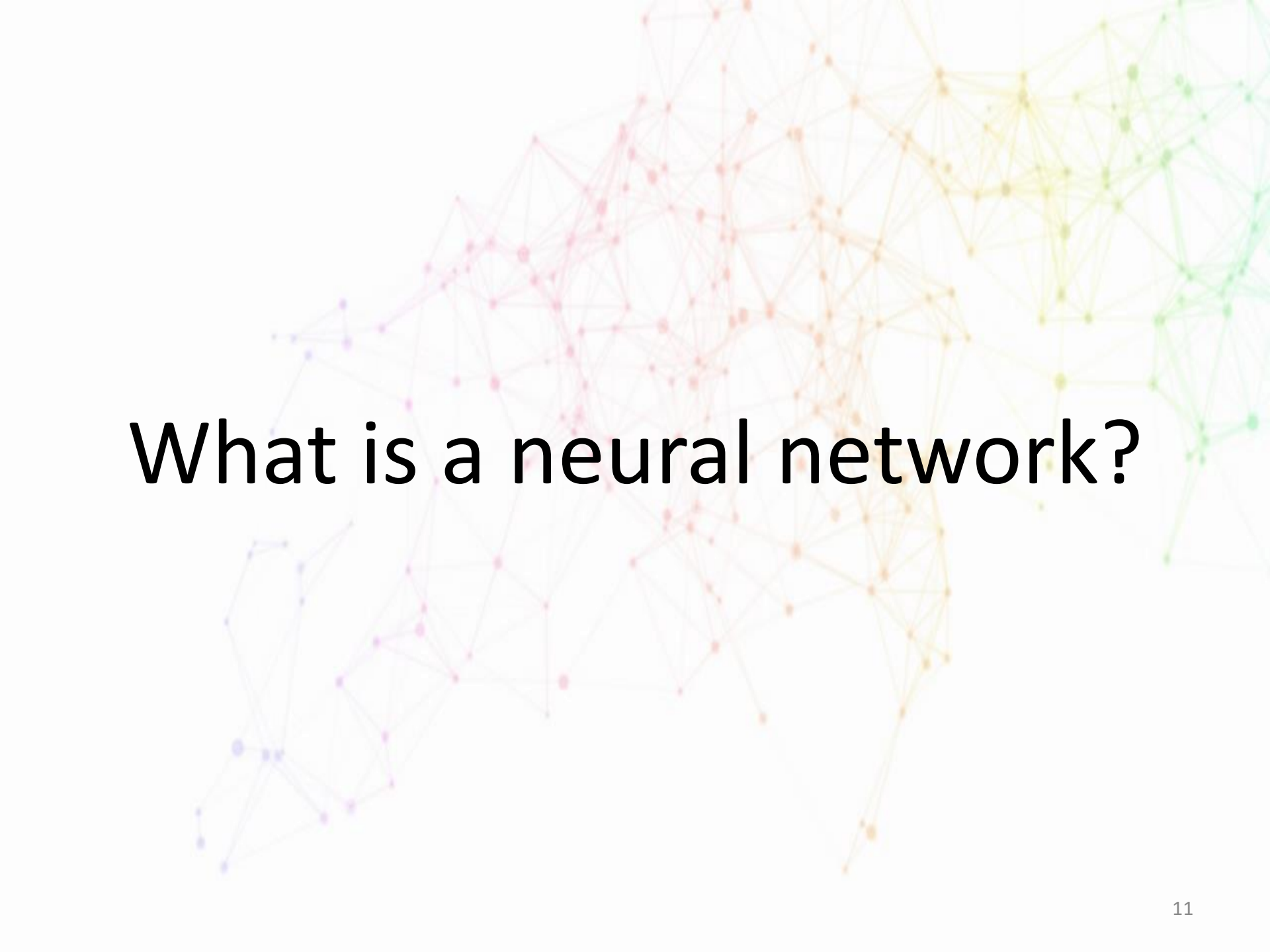
Acknowledgements



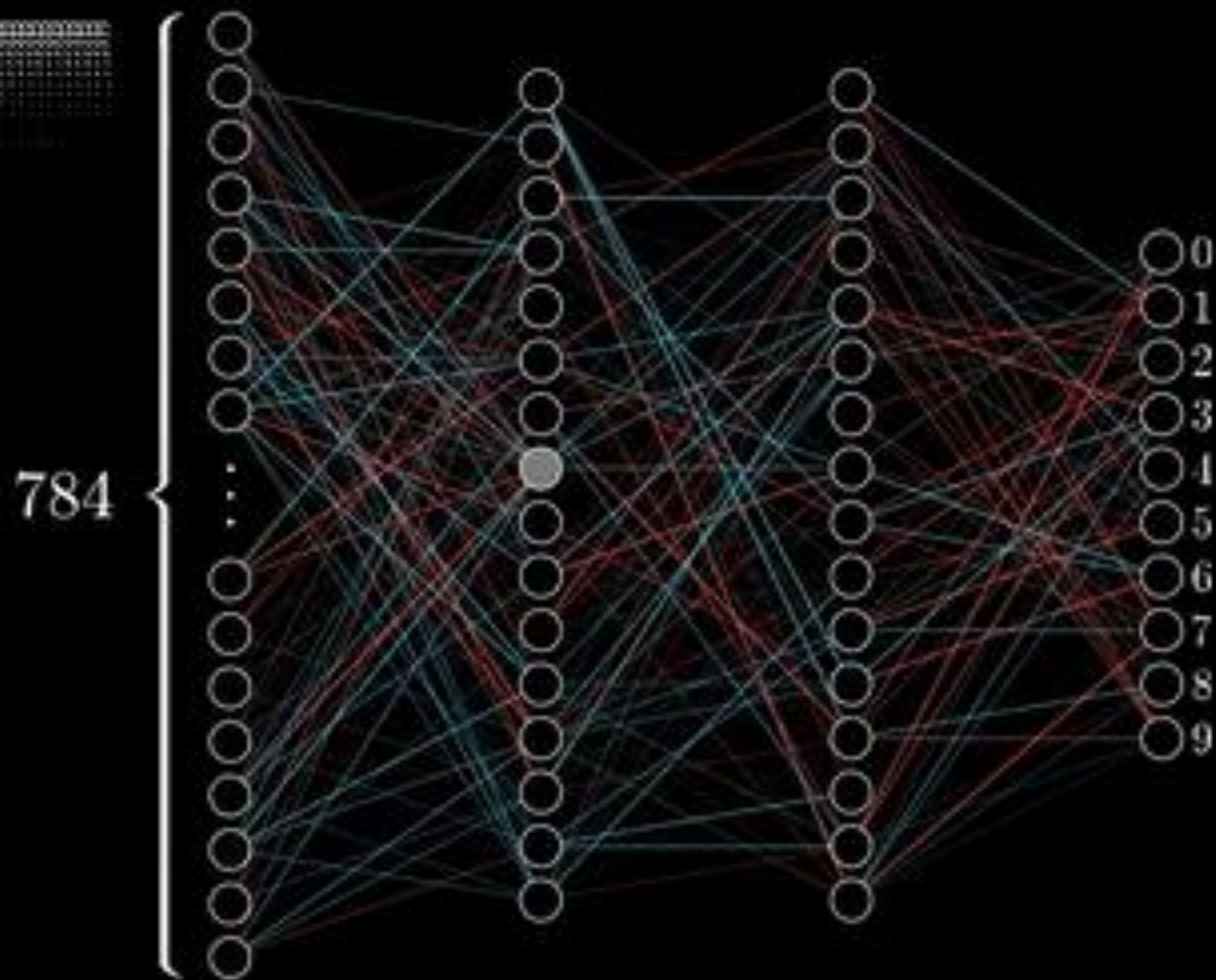
MSRP-Bio Program, Mandana, Paloma, Fiete Lab and my mentor, Daoyuan Qian



Extra slides

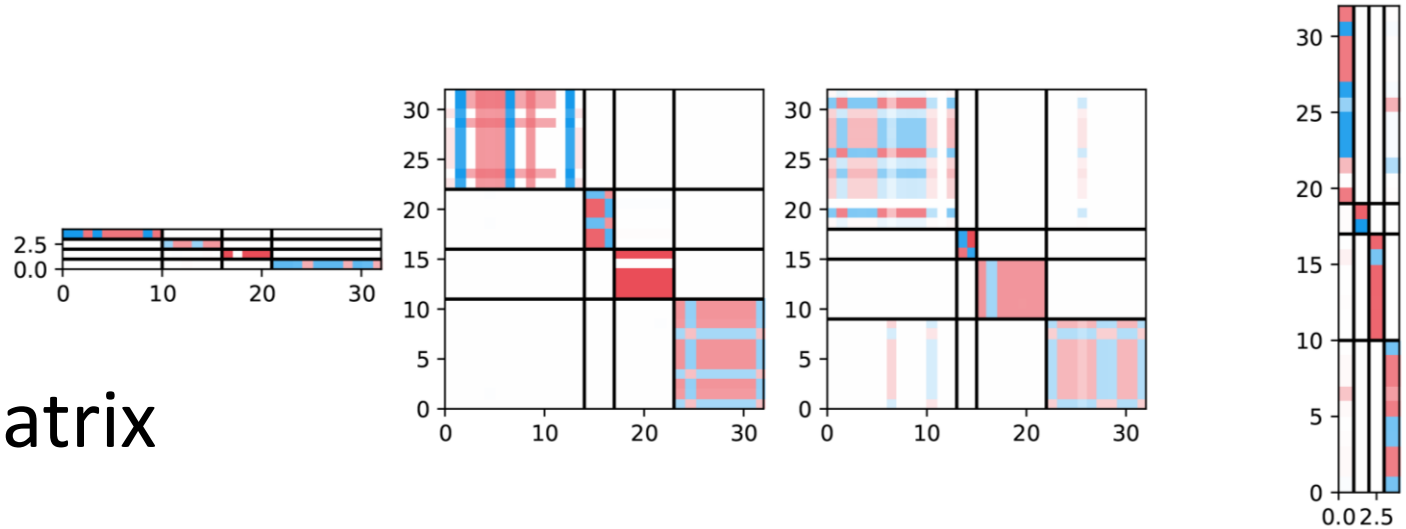


What is a neural network?



Connectivity Matrix Examples

Modular matrix



$$4 \times 32 \times 32 \times 32 \times 4$$

Non-modular matrix

